

Dissecting Executable Scientific Intelligence with Controlled World-Model Interventions

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Abstract

Scientific agents enter experiments with an initial world model that can guide useful search but can also be wrong. Endpoint success alone cannot distinguish correct scientific inference from a favorable heuristic trajectory. We organized a completed three-part programme around three questions: do priors change search and endpoints; does evidence selectively repair wrong priors and yield executable regularities; and can learned information support an unseen action? In a prospective cohort of 135 sessions and 1,243/1,260 planned experiments, prior effects were strongly task dependent. Correct entity information produced a durable advantage in partition, whereas correct structural information gave crystallization an initial head start that narrowed during exploration. Structural partition showed that explicit organization and correctness can contribute separately. Prediction error generally decreased, but selective correction failed at the entity ($p = 0.990$), parametric ($p = 0.079$) and structural ($p = 1.000$) loci. In matched-evidence studies, the structural prediction-update contrast was positive but mixed (+0.0645; 3/5 worlds), while 0/5 summaries recovered the prespecified power law. Executable laws were better than final explicit predictions in only 50/135 cells and worse in 84/135. A formal multi-task open-action assay completed 45/45 scheduled cell records and 240/240 truth plus exact-replay queries; among 42 eligible cells, 11 selected the true top-ranked plan, with 0/42 combining an adequate law and a correct action. These results separate evidence acquisition, numerical revision, structural-law identification, law compression and unseen-action selection. Private replication and context-reset artifact portability are future studies, not results of the present paper.

1. Introduction

Scientific discovery is not simply the production of a high-scoring outcome. A useful result can arise from a correct initial model, a productive but incorrect heuristic, evidence-driven revision or a fortunate endpoint without a reusable account of the relation that generated it. These explanations are scientifically distinct even when the final score is identical. The object of study is therefore the agent’s initial model of what exists, how it works, which quantities matter and where a learned relation should apply.

AI systems can now plan experiments, call chemistry tools and participate in closed-loop discovery workflows [4–6, 17–19]. Interactive scientific environments likewise test hypothesis formation, intervention and inference [1, 7, 9, 11, 21, 22]. Yet endpoint optimization, belief statements, evidence selection, law recovery and action transfer are usually entangled.

Executable chemical worlds provide a controlled instrument for separating these transitions. Entities, process structure, parameters, measurement mappings and private laws can be instantiated while the public task, operations, resource opportunities and external world remain fixed [16]. The present study uses that programmability to intervene on one layer of an initial world model at a time and to score the resulting transitions. The platform qualification is not treated as evidence about agent intelligence.

The paper is organized around three questions.

1. **Do priors change search and endpoints?** We compare opaque, aligned and misspecified initial models within matched task-world clusters and measure both the first action and the best outcome.
2. **Does evidence selectively repair wrong priors and yield executable regularities?** We score held-out predictions, matched-evidence updates and executable law summaries separately from self-reported confidence.
3. **Can learned information support an unseen action?** We distinguish replay of an observed incumbent from ranking fully specified candidate plans whose outcomes were hidden during learning.

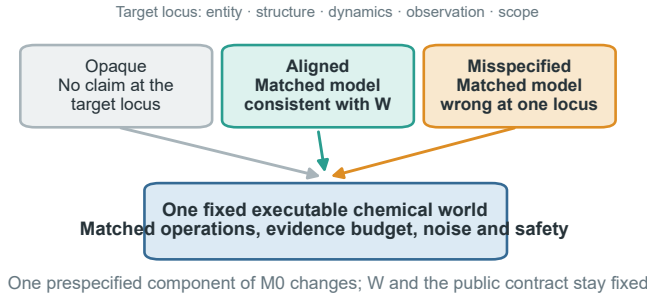
The completed evidence comprises a prospective cohort, a valid matched-evidence follow-up and a formal open-action assay. Early exploratory cohorts, a one-world parametric pilot, an incomplete prototype, repair trajectories and operational accounting are retained in the Supplementary Information. Private replication and context-reset artifact portability are future studies.

2. Related work

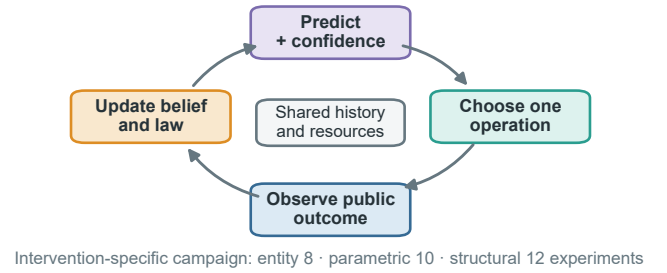
Autonomous chemistry systems demonstrate planning, synthesis, materials search and closed-loop experimentation on real equipment [4–6, 17, 18]. Virtual laboratories and process environments make experimentation repeatable and inexpensive [2, 3, 8, 10, 13]. Interactive discovery

A useful endpoint is not sufficient evidence of law discovery

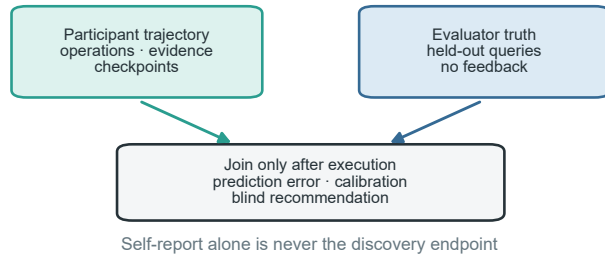
a Intervene on the initial model, not the world



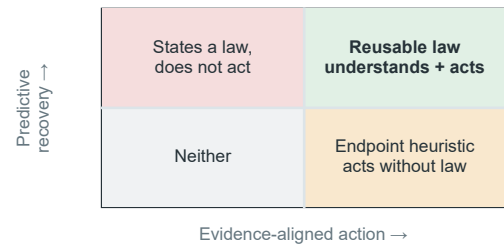
b One session controls a complete campaign



c Score understanding after the campaign



d Classify what the agent actually achieved



The agent-facing initial model is manipulated under one fixed world; prediction, executable law, action and transfer remain separate evidence channels.

Figure 1. The capability chain tested in this paper. **a**, A fixed executable world is paired with opaque, aligned or misspecified information at one declared layer. **b**, A persistent agent selects experiments, observes outcomes and updates predictions and a law summary. **c**, Participant trajectories and evaluator-owned held-out truth remain separate until scoring. **d**, Search, prediction, law recovery and unseen-action selection are distinct outcomes; context-reset portability is a future test.

benchmarks emphasize repeated cycles of hypothesis, intervention and inference [7, 9, 11, 15].

Prior-availability studies and counterfactual-world benchmarks make important contributions, but they often change task information without separating correctness, evidence acquisition and transfer. Mechanism-oriented systems instead optimize model discovery or experiment selection [12, 14, 20]. Our design is complementary: the external world and action space are fixed within matched clusters, while one declared component of the initial scientific model changes. The resulting estimand is a conversion loss along a capability chain, not a leaderboard score.

3. Conceptual framework

Each matched comparison contains one evaluator-owned world $W = (\mathcal{E}, G, \Theta, O, C)$ and one participant-facing initial model $M_0 = (\hat{\mathcal{E}}, \hat{G}, \hat{\Theta}, \hat{O}, \hat{S})$. The public task, action space, observation channels, resource card, safety rules and stochastic identity are fixed. Only one declared component of M_0 changes. The manipulated layer can be entity/ontology, parametric/dynamical or structural/mechanistic; observation and scope are reserved for separate boundary studies.

Within a layer, information is either opaque, aligned with the fixed world or equally detailed but misspecified.

A misspecified model is a plausible scientific representation, not a trick question about obedience. Experimental evidence remains authoritative in all conditions.

We distinguish five outcomes:

1. endpoint optimization;
2. predictive recovery on held-out counterfactual queries;
3. selective correction of a wrong prior;
4. recovery of an executable law that preserves conditional prediction quality; and
5. action transfer to a previously unseen, fully specified plan.

The analysis reports transition losses rather than a composite intelligence score. Endpoint success without predictive and transfer validity is classified as local optimization rather than law discovery.

4. Completed study architecture

The prospective cohort spans five task families with five public worlds per task for the entity layer, and two validated task families with five worlds per task for the parametric and structural layers. Every matched cluster contains opaque, aligned and misspecified arms. Exploratory and prospective world instances are disjoint. Observation-

model interventions and scope/compositional studies are kept outside the current denominator.

The completed programme has three evidence blocks:

1. **Prospective search cohort.** Campaigns contain eight, ten or twelve experiments depending on the intervention layer, with checkpoints fixed before execution. Each checkpoint records predictions, uncertainty, evidence references, a typed law summary and the next experimental intent.
2. **Matched-evidence follow-up.** The same contradictory evidence is presented to each arm after the initial search. The valid analysis combines an unaffected parametric block and a corrected structural phase-process block; a superseded run is retained only in the Supplementary Information because its evaluator omitted the declared intervention.
3. **Formal open-action assay.** After twelve experiments, the agent ranks eight fully specified candidate plans with outcomes hidden. Public, truth-evaluated and executed plans are verified as identical. This assay addresses unseen-action transfer rather than replay of an observed incumbent.

5. Results I — Priors change search and endpoints

The prospective cohort completed **135/135** scheduled sessions and **1,243/1,260** planned experiments. The denominator contains 1,269 closed batch lifecycles, 1,243 final assays and 26 discarded lifecycles. Failed cells and uncommitted attempts remain in their assigned denominators.

Prior effects were task dependent. In entity-level partition, aligned information improved the first experiment by **0.106** and the best observed endpoint by **0.200** relative to misspecified information, with the same direction in **5/5** worlds. In structural crystallization, aligned information improved the first experiment by **0.141** in **5/5** worlds, but the best-endpoint gap narrowed to **0.055** and was positive in only **3/5** worlds. Structural partition separated organization from correctness: aligned and misspecified models exceeded opaque identifiers by **0.163** and **0.143**, respectively, while their mutual difference was only **0.020**.

The remaining task-locus combinations bounded these cases. Entity-level reaction safety showed a small aligned-minus-misspecified difference of **0.036** at the best endpoint, distillation narrowed from **0.053** on the first experiment to **0.011** at the best endpoint, and electrochemical and crystallization effects were heterogeneous. Correct information was therefore neither a universal performance intervention nor a necessary condition for useful search.

The agents did perform substantive within-session search. **91.2%** of completed experiments used a unique recipe, **84.4%** of session optima occurred after the campaign midpoint and **32.6%** occurred in the final completed experiment. These observations support search adaptation, not scientific correction: a productive trajectory can arise while the initial model remains wrong.

6. Results II — Evidence, selective correction and executable laws

All sessions submitted five belief checkpoints, yielding **675/675** typed records and **6,300** prespecified counterfactual predictions. The evaluator completed **420/420** truth executions and scored every checkpoint. Prediction error generally decreased, but the registered selective-correction criterion failed at all three loci: entity $p = 0.990$, parametric $p = 0.079$ and structural $p = 1.000$. General prediction improvement and targeted repair of an incorrect prior were therefore not interchangeable outcomes.

Matched evidence localized the loss without reducing it to a binary label. In the parametric block, all five misspecified summaries rejected the supplied high-potential direction after receiving the same contradictory evidence. In the structural block, the misspecified-minus-aligned prediction-update contrast was **+0.0645**, positive in **3/5** worlds (exact one-sided sign-flip $p = 0.125$), while **0/5** misspecified summaries recovered the prespecified power law. Evidence acquisition, numerical revision and structural-law identification therefore remain separable transitions.

The evaluator executed all **135/135** final typed laws. Laws were more accurate than final explicit predictions in **50/135** cells, equal in one and worse in **84/135**; mean law-minus-final error was **+0.069**. Syntax and execution were reliable, but compression into a reusable relation was often lossy. Blind replay of the committed recommendation versus the observed incumbent was better, equivalent and worse in **1/119/1** of **121** evaluable cells. Reproducibility of an observed action was therefore almost complete, while improvement beyond the observed history was rare.

7. Results III — Learned information and unseen action

The formal open-action assay completed **45/45** scheduled cell records. Independent evaluation completed **240/240** truth executions and **240/240** exact replays, and verified identity among the public, truth-evaluated and executed action plans. **42/45** cells were eligible for action metrics; three crystallization failures remain in the scheduled denominator.

Among eligible cells, **11/42** selected the true Top-1 plan, with mean selected rank **3.31/8** and mean normalized regret **0.297**. The joint mechanism-action outcome was more informative than Top-1 alone: **30/42** cells had an inadequate law and a wrong action, **11/42** had an inadequate law but a correct action, **1/42** had an adequate law but a wrong action and **0/42** combined an adequate law with a correct action. Law adequacy was therefore not sufficient for action correctness, while a correct action could occasionally occur without an adequate law.

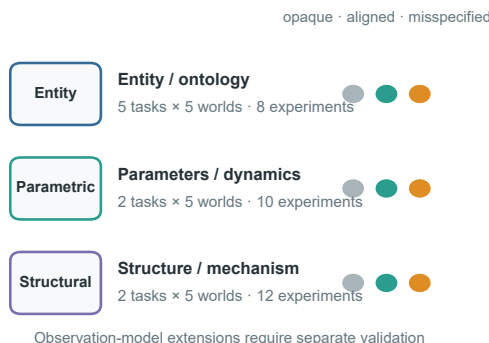
Task heterogeneity was substantial. Electrochemical conversion reached Top-1 in **4/15** cells, reaction safety in **4/15** and crystallization in **3/12** eligible cells. The assay is descriptive: only twelve task-world clusters retain complete three-arm comparisons, and failures are concentrated in crystallization. The bounded conclusion is a transfer

A sparse programme intervenes on the initial world model one locus at a time

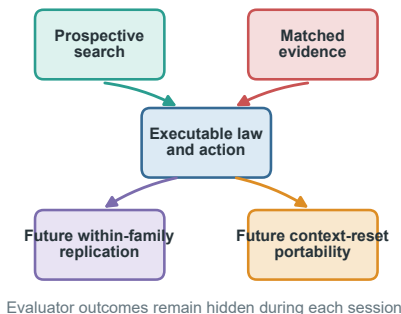
a Five task families form the entity / ontology backbone



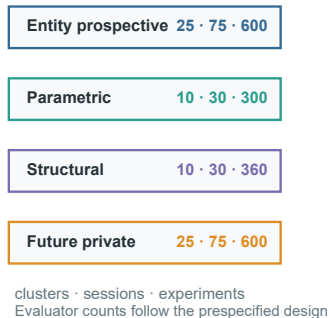
b Prospective search blocks



c Evidence partitions



d Planned block denominators



World programmability defines the intervention space; validation, matched arms and separate denominators preserve causal interpretability.

Figure 2. Completed evidence architecture. **a**, The entity backbone contains five task families and five public worlds per task. **b**, Each matched task-world cluster contains opaque, aligned and misspecified initial models for one declared locus. **c**, Parametric and structural blocks are validated separately. **d**, Prospective search, matched evidence, evaluator scoring and open-action tests retain separate sessions and denominators; private replication and context-reset portability are future work.

boundary, not an arm-level effect.

8. Discussion

The completed evidence supports three bounded conclusions. First, priors change search and endpoints, but their effects are context dependent: a correct model can provide a durable advantage, an early head start or no stable benefit, and a wrong model can still guide productive exploration. Second, prediction improvement does not establish selective correction. Matched evidence can trigger evidence acquisition and numerical revision without reliable structural-law identification, and executable summaries can lose information relative to the agent’s conditional predictions. Third, even a complete action interface does not guarantee transfer to an unseen candidate set: only 11/42 eligible readouts selected the true Top-1 plan, and no cell combined an adequate law with a correct action.

These results argue against a single endpoint score for scientific intelligence. Evidence acquisition, belief revision, law identification, law compression and action transfer should be measured as separate transitions. The correct interpretation is a map of conversion losses in one controlled agent-system configuration, not a claim about all

models or all chemical settings.

8.1 Scope and limitations

The study uses bounded executable chemical worlds rather than universal chemical fidelity or direct wet-laboratory validation. Five independent worlds per task limit precision for small or heterogeneous effects. The observation-model intervention and scope/compositionality intervention were not executed. The matched-evidence structural block contains five worlds, so its sign-flip result is descriptive. The open-action assay contains 42 eligible cells, three retained crystallization failures and twelve complete three-arm clusters; its arm summaries are not causal estimates. Exact replay verifies the declared computational world and plan semantics but does not remove all interface or execution burden.

9. Methods

9.1 World and initial-model construction

Each task instantiates an executable world and a participant-facing initial model. Within a matched cluster, all arms share the world, resource card, public contract and stochastic identity; exactly one declared component of the

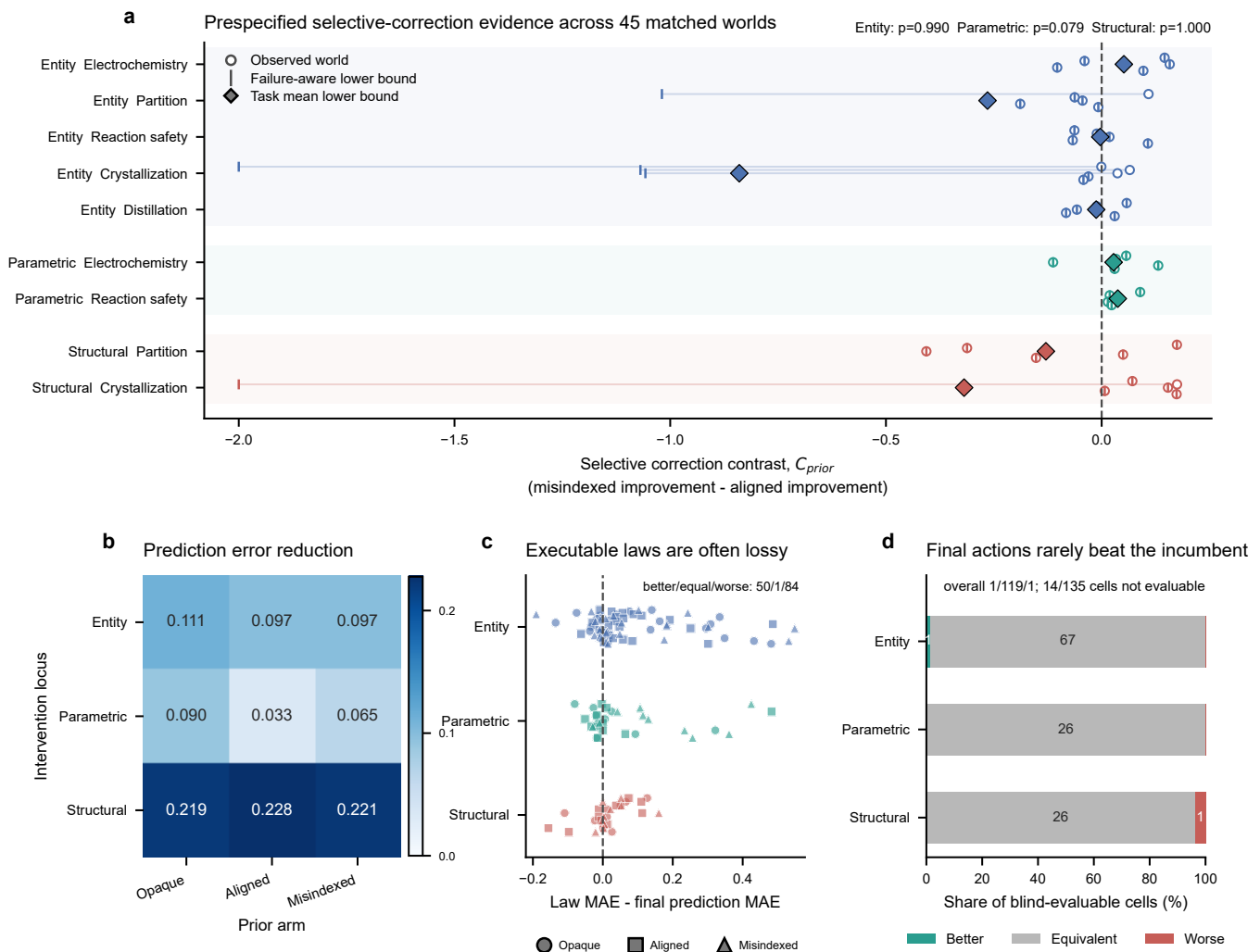


Figure 3. Prediction, law and action dissociation. Held-out prediction error, selective-correction contrasts, executable-law error and blind incumbent replay are separate readouts. The completed cohort shows broad prediction improvement but no passed selective-correction gate, frequent lossy law compression and almost no recommendation gain over the observed incumbent.

initial model changes. Entity, parametric and structural encodings are included only after separate identifiability checks confirm comparable information volume, wording, confidence and falsification cost.

9.2 Campaigns, checkpoints and evaluator scoring

Campaign lengths and checkpoint positions were fixed before execution. A checkpoint records prior assessment, predictions, uncertainty, evidence references, an executable law summary and next-experiment intent. The evaluator owns the held-out truth queries and does not return them to the participant. Prediction error is the mean normalized absolute error across prespecified query-metric pairs. Final laws are executed on the same coordinates, and blind replay compares the committed recommendation with the observed incumbent using separate evaluator resources.

9.3 Matched evidence and selective-correction estimand

For locus ℓ , the primary contrast is

$$C_{\ell} = (E_{\text{mis,pre}} - E_{\text{mis,final}}) - (E_{\text{aligned,pre}} - E_{\text{aligned,final}}).$$

The selective-correction gate requires improvement in the misspecified arm, no unacceptable degradation in the aligned arm and a lower confidence bound above zero. Loci are reported separately; unlike endpoint contrasts are not pooled as one universal prior effect. Failed scientific cells remain in the denominator and are not replaced.

9.4 Open-action construction and analysis

An outcome-blind generator constructed eight complete candidate action plans per world before participant execution. Each plan disclosed its full operation sequence, parameters, initial-state assumptions, measurements and terminal assay. After twelve experiments and the final checkpoint, the participant returned a complete ranking and one selected plan. The evaluator then calculated selected rank, Top-1, regret and the joint law-action category. All scheduled cells remained in the denominator.

Formal open-action matrix: complete plans still do not close law-to-action transfer

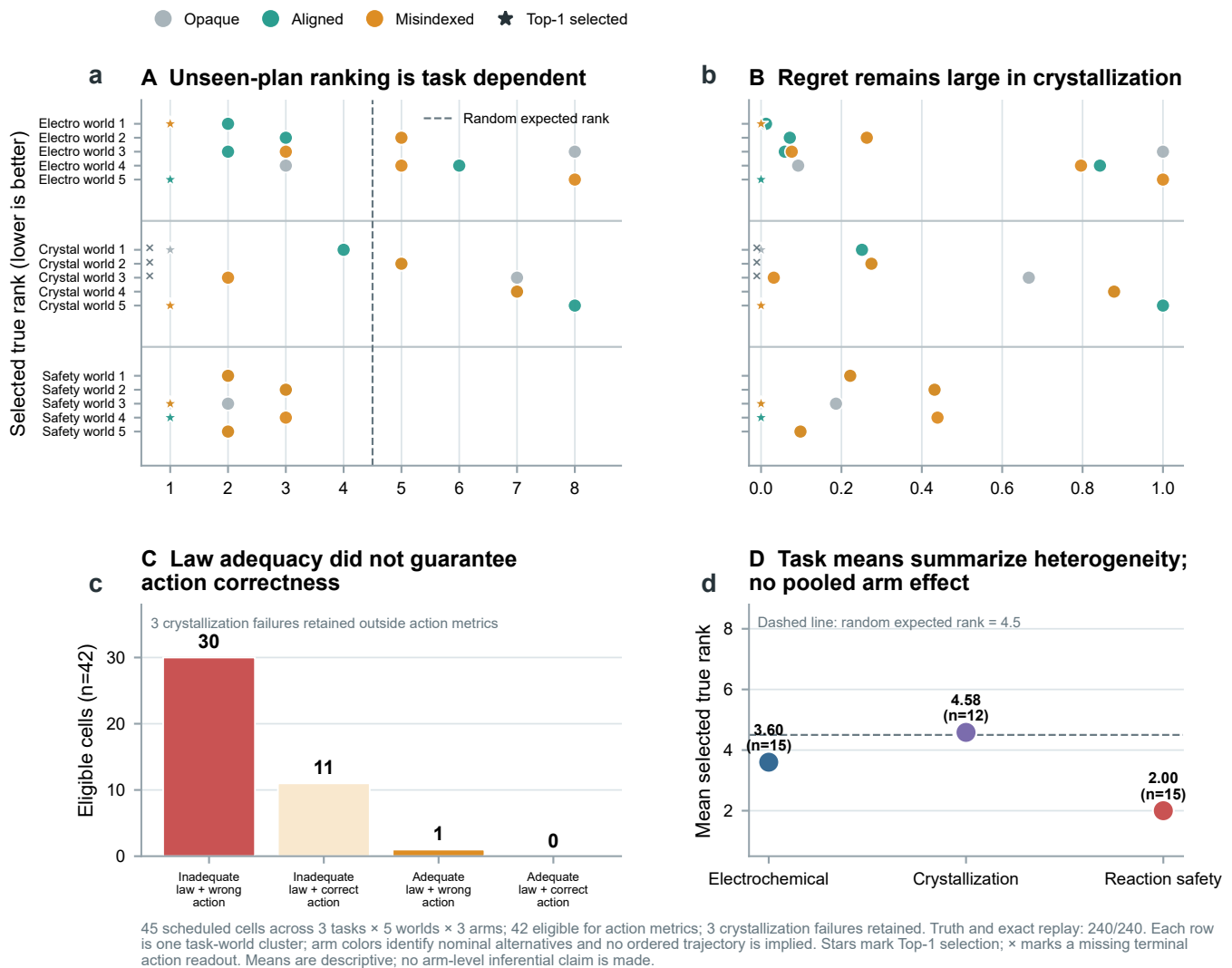


Figure 4. Unseen-action selection after exploration. Selected rank, normalized regret, joint law–action categories and task-level means are shown for the 42 eligible cells. The three crystallization failures remain in the scheduled denominator. Every candidate plan was fully specified and executed exactly as disclosed; candidate outcomes and ranks were hidden until the terminal readout.

10. Completed studies and future programme

The following boundary is part of the paper’s interpretation.

- **Completed — prospective cohort:** priors change search and endpoints; selective correction is not established.
- **Completed — matched-evidence follow-up:** evidence acquisition and numerical revision can occur without exact structural-law recovery.
- **Completed — formal open-action assay:** unseen-action transfer is limited and task dependent.
- **Supplementary — exploratory cohorts and pilot:** developmental context, not pooled into the main denominators.

- **Supplementary — incomplete prototype, repair trajectory and excluded outputs:** diagnostic history; never used to replace a scheduled cell.
- **Future — private within-family replication:** tests stability on newly sealed worlds.
- **Future — context-reset artifact portability:** tests whether learned artifacts transfer to a new target context.

The future studies require new protocols and independent denominators. They are not implied by the current endpoint results and are not described as completed evidence.

11. Data and code availability

The executable environment, prespecified protocols, analysis code, source data and reproducible figure scripts accompany the public release. Raw model payloads, credentials and private world instances are excluded. Future private and context-reset studies will use newly sealed instances and will not alter the present denominators.

12. Competing interests

The authors declare no competing interests.

Supplementary Information

S1. Early exploratory cohorts

Before the prospective cohort, exploratory configurations were used to test whether explicit priors altered endpoint behavior and whether the evaluator could score prediction, law and blind replay without changing the participant trajectory. These cohorts reached 44/45 and 75/75 terminal cell records in their respective configurations, but their task mix, recovery rules and campaign lengths were not identical to the prospective design. They are therefore descriptive context only. Across the common endpoint panels, explicit information sometimes improved outcomes over opaque identifiers, while misspecified information was not consistently harmful. Held-out prediction error often fell, but the wrong-prior improvement was not selectively larger and blind replay rarely exceeded the observed incumbent.

S2. One-world parametric pilot

A preliminary one-world parametric pilot retained one opaque, one aligned and one misspecified cell. All 3/3 cells, 12/12 experiments, 12/12 checkpoints, 4/4 shared truth queries and 18/18 blind replays completed with exact replay. The misspecified agent moved outside its supplied parameter window after contradictory evidence and reduced its stated reliability, yet its best endpoint remained below the aligned and opaque cells. The pilot demonstrates that model rejection and finite-budget policy recovery are distinct; it is not a cross-task claim.

S3. Incomplete prototype and repair trajectory

An earlier open-action prototype retained 11/15 complete cells and did not produce a Top-1 selection. It is not combined with the formal assay. A separate repair trajectory completed a previously interrupted crystallization cell but included a resource rejection and selected rank 8/8. The repair is a sensitivity result, not a replacement cell.

S4. Operational diagnostics and excluded outputs

The development records contain schema failures, rejected operations, recovery events, resource rejections, timing summaries and other interface diagnostics. These quantities describe the complete agent system and are not independent scientific samples. Implementation counters, run identifiers, cell identifiers and raw or replaced outputs are intentionally omitted from the reader-facing paper.

Historical outputs remain archived for auditability but cannot alter a completed denominator.

S5. Future studies

Private replication will use newly sealed worlds with the same three-arm contract and one-shot denominators. Context-reset portability will compare raw evidence, structured evidence bundles and typed executable laws in a new target context. Both studies require independent preregistration and will be reported separately from the present paper.

References

- [1] Serafim Batzoglou. ReplaySCM: A benchmark for executable causal mechanism induction from interventions, 2026. Preprint.
- [2] Chris Beeler, Sriram Ganapathi Subramanian, Kyle Sprague, et al. ChemGymRL: A customizable interactive framework for reinforcement learning for digital chemistry. *Digital Discovery*, 3:742–758, 2024. doi: 10.1039/D3DD00183K.
- [3] Maximilian Bloor, José Torracca, Ilya Orson Sandoval, et al. PC-Gym: Benchmark environments for process control problems, 2024. Preprint, arXiv:2410.22093.
- [4] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.
- [5] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6:525–535, 2024. doi: 10.1038/s42256-024-00832-8.
- [6] Kourosh Darvish, Marta Skreta, Yuchi Zhao, et al. OR-GANA: A robotic assistant for automated chemistry experimentation and characterization. *Matter*, 8(2), 2025. doi: 10.1016/j.matt.2024.10.015.
- [7] Haonan Duan, Stephen Zhewen Lu, Caitlin Fiona Harrihan, Nishkrit Desai, Jiarui Lu, Michał Koziarski, Leonardo Cotta, and Chris J. Maddison. Measuring scientific capabilities of language models with a systems biology dry lab, 2025. Preprint, arXiv:2507.02083.
- [8] Kobi C. Felton, Jan G. Rittig, and Alexei A. Lapkin. Summit: Benchmarking machine learning methods for reaction optimisation. *Chemistry-Methods*, 1:116–122, 2021. doi: 10.1002/cmt.202000051.
- [9] Kanishk Gandhi, Michael Y. Li, Lyle Goodyear, Agam Bhatia, Louise Li, Aditi Bhaskar, Mohammed Zaman, and Noah D. Goodman. BoxingGym: Benchmarking progress in automated experimental design and model discovery, 2025. Preprint, arXiv:2501.01540.
- [10] Florian Häse, Matteo Aldeghi, Riley J. Hickman, Loïc M. Roch, Melodie Christensen, Elena Liles, Jason E. Hein, and Alán Aspuru-Guzik. Olympus: a benchmarking framework for noisy optimization and experiment planning. *Machine*

- Learning: Science and Technology*, 2:035021, 2021. doi: 10.1088/2632-2153/abedc8.
- [11] Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0324.
- [12] Sanchit Kabra, Nikhil Abhyankar, Saaketh Desai, Prasad Iyer, and Chandan K. Reddy. LLM-AutoSciLab: Closed-loop scientific discovery via active experimentation with llms, 2026. Preprint, arXiv:2605.24043.
- [13] Shreshth A. Malik, Tiarnan Doherty, Panagiotis Tigas, Muhammed Razzak, Stephen J. Roberts, Aron Walsh, and Yarin Gal. MADE: Benchmark environments for closed-loop materials discovery. In *International Conference on Machine Learning*, 2026.
- [14] Kevin Murphy. Model discovery agent: Llm-assisted bayesian experiment design for data-efficient discovery of mechanistic world models, 2026. Preprint, arXiv:2608.09696.
- [15] Maximilian Nägele and Florian Marquardt. Agentic exploration of physics models. *Physical Review X*, 16:031002, 2026. doi: 10.1103/xnqc-q6nt.
- [16] Jiangjie Qiu, Yijun Li, and Xiaonan Wang. ChemWorld: Programmable chemical worlds for controlled and replayable agent experimentation, 2026. Preprint, arXiv:2608.10792.
- [17] Tao Song, Man Luo, Xiaolong Zhang, et al. A multiagent-driven robotic AI chemist enabling autonomous chemical research on demand. *Journal of the American Chemical Society*, 147(15):12534–12545, 2025. doi: 10.1021/jacs.4c17738.
- [18] Nathan J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624:86–91, 2023. doi: 10.1038/s41586-023-06734-w.
- [19] Aikaterini Vriza, Michael H. Prince, Tao Zhou, Henry Chan, and Mathew J. Cherukara. Operating advanced scientific instruments with AI agents that learn on the job. *npj Computational Materials*, 12:160, 2026. doi: 10.1038/s41524-026-02005-0.
- [20] Stefan Wahl, Raphaela Schenk, Ali Farnoud, Jakob H. Macke, and Daniel Gedon. A probabilistic framework for llm-based model discovery, 2026. Preprint, arXiv:2602.18266.
- [21] Junlin Yang, Dylan Zhang, Xiangchen Song, et al. CausaLab: A scalable environment for interactive causal discovery toward ai scientists, 2026. Preprint.
- [22] Tianshi Zheng, Kelvin Kiu-Wai Tam, Newt Hue-Nam K. Nguyen, et al. NewtonBench: Benchmarking generalizable scientific law discovery in llm agents. In *International Conference on Learning Representations*, 2026.